

mechanisms to manage disk access that filesystem drivers use. LVM doesn't change the data when you store it; it only manages where it is stored on the disk. You certainly won't feel a difference in performance if you use LVM. If you measure it, you'll find that the performance drop is negligible, and in most cases, immeasurable.

Getting started with LVM

LVM has two types of volumes you need to be concerned about - physical and logical volumes. Physical volumes are the actual disks (or partitions, loop files or any block device) that store your LVM set. Logical volumes are the volumes that you actually mount and store your data in.

In between the physical and the logical volumes lies the volume group. You create a volume group on one or more physical volumes, and then you create logical volumes inside a volume group.

The additional abstraction of a volume group might not seem very intuitive at first, but consider that you can't resize physical volumes at will. You can, however, resize volume groups in that you can extend them to use an arbitrary number of physical volumes. Logical volumes must reside inside their volume group. So the volume group serves as an elastic container for your volumes.

Now let's get around to using it. To begin using LVM, you need some physical volumes first. You need an old fashioned partition. This is what the partition table on my laptop looks like:

```
Node Size   Code   Name
sda1 512.0 MiB EF00   EFI System
sda2 127.5 GiB 0700   Microsoft basic data
sda3 337.8 GiB 8E00   Linux LVM
```

Yes, since I use UEFI, I have a 512 MB EFI system partition up front. That's FAT32. Then there's a 127.5 GB NTFS partition that holds Windows 8.1 for me. The rest, as you can see, is a LVM partition. 8e00 is its type ID. 8e00 is the partition type for a LVM physical volume. Yes, when I said the days of partitioning are over, I lied a bit. Even if you want to dedicate an entire disk to LVM, you need to create at least one partition on it and mark it 8e00 (or just 8e if you're using MBR partitions).

So now that I've created a partition for my LVM physical volume, how do I use it? Well, I need to initialise it first (format it with an LVM, if you will), and to do that, I type in the following command:

```
$: sudo pvcreate /dev/sda3
```

...replacing sda3 with whatever my node is. If you're using a SSD and want to make sure your blocks are aligned to 1MB, you might want to run the following command instead:

```
$: sudo pvcreate --dataalignment 1m /dev/sda3
```

And we've just created a physical volume. I now run:

```
$: sudo pvdisplay
```

And the system gives me a ton of information about my newly created physical volume. Running *pvdisplay* on my laptop gives me the following:

```
--- Physical volume ---
PV Name               /dev/sda3
VG Name               VolGroup
PV Size                337.76 GiB / not usable 4.01 MiB
Allocatable           yes
PE Size                4.00 MiB
Total PE               86466
Free PE                40386
Allocated PE           46080
PV UUID                AKEk7x-aPzN-yLP1-F6u3-d1gy-wbdn-SNxZF0
```

So now that I have a physical volume to play around with, why not create a volume group on it? Let's say I want a volume group called *VolGroup*, then I run the following command:

```
$: sudo vgcreate VolGroup /dev/sda3
```

My volume group is ready. I can now begin creating my volumes on it, but how do I do that? To create a standard volume, I can use the following command:

```
$: sudo lvcreate -L 64G VolGroup -n Root
```

I just created a 64 GB logical volume in the *VolGroup* volume group and named it 'Root'. It'll now show up as */dev/VolGroup/Root* and */dev/mapper/VolGroup-Root*. I can use whichever node strikes my fancy. Now I can create a filesystem on it and mount it somewhere, like shown below:

```
$: sudo mkfs.ext4 /dev/VolGroup/Root
$: sudo mount /dev/VolGroup/Root /mnt
```

If I wanted a swap partition inside my volume group, I could just create a standard volume, but it might be a good choice to mark the volume as contiguous, so that the volume is stored on concurrent blocks on a single physical volume. This takes away a lot of the flexibility offered by LVM, but to minimise disk access time, this is a good option. Let's run the following command:

```
$: sudo lvcreate -C y -L 4G VolGroup -n Swap
```